Instruction Tuning and Direct Preference Optimization on a Small Language Model

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Repository: https://github.com/mabdullahi7780/small-model-alignment-starter

Summary. We fine-tune `Qwen/Qwen2.5-0.5B` on 30,000 single-turn UltraChat [7] pairs (SFT) and then train two DPO policies from the SFT checkpoint, one on 15,000 HH-RLHF [4] pairs and one on 15,000 UltraFeedback [8] pairs, following [2]. Separately, we train a rank-16 LoRA [5] adapter on `SmolLM2-360M-Instruct` with TRL's `DPOTrainer` on the provided persona pairs. Under a blind pairwise judge (`google/gemini-3.6-flash`), HH-DPO beats SFT on the HH safety set (300 items, 185 distinct prompts) with a 67.5% win rate excluding ties, mostly by redirecting harmful requests rather than refusing them. UltraFeedback-DPO does not beat SFT on 300 AlpacaEval prompts (44.8%); it loses mainly where its answer is the longer one (41.1%, against 68.8% when it is not). The LoRA persona wins every decided style comparison (24-0) and is judged less useful than its base (13-25), although with 40 prompts this is not conclusive (95% CI 21.2 to 50.1%).

1 Repository, Models, and Data

Repository and code. Training and evaluation code is in `src/pa2/` and is exposed to the grader through the root-level `submission_adapter.py`. Run settings are in `configs/`, and each task has one Kaggle notebook, `notebooks/01_setup.ipynb` to `notebooks/07_judge.ipynb`. The SFT and both DPO training loops are written directly in PyTorch; Transformers is used only to load models and tokenizers. TRL [10] is used only in `src/pa2/lora_persona.py`; PEFT is used there and in `src/pa2/generate.py`, which loads the adapter for generation. A unit test checks that no other training code imports TRL or a prebuilt Trainer.

Models. `Qwen/Qwen2.5-0.5B` (494M parameters) is the base for SFT and both DPO runs. `HuggingFaceTB/SmolLM2-360M-Instruct` (362M parameters) is the base for the LoRA persona run.

Table 1: Training data for each track. Revisions are Hugging Face Hub commits. All sampling uses seed 42, except the UltraFeedback training draw, which uses seed 43 so that it is a different shuffle from validation.

Track	Source (revision)	Filtering	Train	Val.
SFT	`HuggingFaceH4/ultrachat_200k` (`8049631`), `train_sft` / `test_sft`	First user turn and first reply only. Kept only if the formatted prompt, response and EOS fit in 512 tokens (51,867 rows scanned: 21,862 too long, 5 not a user-then-assistant turn). Evaluation-set and repeated prompts are filtered; none occurred.	30,000	1,000
HH-DPO	`Anthropic/hh-rlhf` (`09be8c5`), `harmless-base` and `helpful-base`	Single-turn rows only (71% of scanned training rows were not clean single-turn pairs); identical prompt in chosen and rejected; prompt at most 256 tokens, prompt plus each response plus EOS at most 512; evaluation-set and repeated prompts removed. Balanced halves.	7,500 + 7,500	500 + 500
UltraFeedback-DPO	`HuggingFaceH4/ultrafeedback_binarized` (`3949bf5`), `train_prefs`	Ties dropped (7,387 of 61,135 rows); same length limits as HH; evaluation-set and repeated prompts removed; validation drawn disjoint from training.	15,000	1,000
LoRA persona	`data/persona_lora/ramsay_style_preferences.json`	Seeded split of the 220 provided pairs.	200	20

Evaluation sets. **HH-300:** the provided `eval_sets/hh_rlhf_eval_300_seed42.json`, 300 synthetic items in 10 categories of 30, each annotated with its expected behavior. The file holds only 185 distinct prompts: 71 prompts appear 2 to 6 times, and the 30 self-harm items are 5 prompts repeated 6 times. With greedy decoding both models answer every copy identically in 63 of the 71 repeated groups, so repeats are not independent (Section 5). **AlpacaEval-300:** 300 distinct prompts out of the 805 in `tatsu-lab/alpaca_eval` (revision `2edc6fa`), drawn with `sorted(random.Random(42).sample(range(805), 300))` and stored as `eval_sets/alpaca_eval_300_seed42.json`. **Persona-40:** 40 distinct style prompts written for this assignment, `eval_sets/persona_style_eval_40.json`, with 10 each of cooking, coding, general help and safety-sensitive requests; the largest Jaccard word overlap with any training prompt is 0.29.

Prompt formatting and seeds. All Qwen training and generation uses the template

```
<|im_start|>user\n{instruction}<|im_end|>\n<|im_start|>assistant\n
```

with no system prompt, and every response ends with the tokenizer EOS `<|endoftext|>`. The SmolLM2 runs use that model's own chat template, which inserts its default system line. Prompt and response are tokenized separately and right-padded for training. Labels and attention masks are built from token *positions*, never token ids, because Qwen's pad token is its EOS token (id 151643): an id-based mask would hide the EOS that every response must learn to produce. All seeds are 42 (43 for the UltraFeedback training draw).

2 Experimental Configuration

2.1 Objectives and metrics

For a prompt x and response y, $\log \pi(y \mid x) = \sum_{t=1}^{|y|} \log \pi(y_t \mid x, y_{<t})$ is summed over response tokens only; prompt and padding positions are masked. The SFT loss is the token mean over the $B = 32$ examples of one optimizer step:

$$\mathcal{L}_{\mathrm{SFT}}(\theta) = -\frac{1}{\sum_{i=1}^{B} |y_i|} \sum_{i=1}^{B} \sum_{t=1}^{|y_i|} \log \pi_\theta(y_{i,t} \mid x_i, y_{i,<t}) \,. \tag{1}$$

For a preference pair (x, y_w, y_l), DPO [1] uses the implicit reward

$$r_\theta(x, y) = \beta \left[\log \pi_\theta(y \mid x) - \log \pi_{\mathrm{ref}}(y \mid x)\right]$$

and minimizes

$$\mathcal{L}_{\mathrm{DPO}}(\theta) = -\,\mathbb{E}_{(x,y_w,y_l)} \left[\log \sigma(r_\theta(x, y_w) - r_\theta(x, y_l))\right], \qquad \beta = 0.1. \tag{2}$$

Since $\pi_\theta = \pi_{\mathrm{ref}}$ at initialization, both rewards are zero and the loss is exactly $\ln 2 \approx 0.6931$, which we use as a correctness check. *Reward accuracy* is the fraction of pairs with $r_\theta(x, y_w) > r_\theta(x, y_l)$ (strict, so ties count as wrong), and *reward margin* is the mean of $r_\theta(x, y_w) - r_\theta(x, y_l)$. For a pairwise comparison with W wins, T ties and L losses we report the win rate with ties counted as half, $(W + T/2)/(W + T + L)$, the win rate excluding ties, $\hat{p} = W/n$ with $n = W + L$, its 95% Wilson score interval ($z = 1.96$)

$$\mathrm{CI}_{95} = \frac{\hat{p} + \frac{z^2}{2n} \pm z\sqrt{\frac{\hat{p}(1-\hat{p})}{n} + \frac{z^2}{4n^2}}}{1 + \frac{z^2}{n}}, \tag{3}$$

and an exact two-sided sign test of W against L. We use the Wilson interval because the simpler Wald interval is unreliable for the small persona set and collapses to a point at 24-0 (the summary files in `results/task7_judge/` also store the Wald interval).

2.2 Run configuration

All runs used Kaggle with two Tesla T4 GPUs (14.56 GiB each), PyTorch 2.11.0+cu128, Transformers 4.57.6, TRL 1.15.0 and PEFT 0.20.0. Each run trains on a single GPU with no data parallelism; the DPO runs place the frozen reference model on the second GPU. Table 2 gives the values actually used for the submitted checkpoints.

Checkpoint selection. SFT: lowest loss on the 1,000 validation pairs, evaluated every 100 steps (best is the final step, 938). DPO: highest validation reward accuracy on 1,000 pairs, evaluated at step 0 and every 50 steps (HH: step 469; UltraFeedback: step 350, whose accuracy of 0.710 against 0.700 at the final step is within one standard error of about 0.014); final checkpoints were kept as well. LoRA: the final adapter after 75 steps.

3 Parameter Choices and Deviations

Learning rate, β, batch size, epochs, maximum length and dataset sizes follow the guide configuration for SFT and both DPO runs. Table 3 lists every change and every choice the guide left open, grouped by whether it was driven by resources, by measured performance, or by design.

4 Training Behavior

SFT. Validation loss fell from 1.369 to 1.318 nats per response token (perplexity 3.93 to 3.74) with no sign of overfitting. It rose slightly, to 1.375 at step 100, while the learning rate was at its peak, and then fell steadily under cosine decay. The best checkpoint is the last one and the loss was still falling, so a longer run would likely

Table 2: Configuration of the four training runs and of evaluation-time generation (values used for the submitted checkpoints).

(a) Parameters and optimization

Run	Trainable parameters	Optimizer	Peak LR and schedule	Weight decay	Clip
SFT	Full, 494M	AdamW (0.9, 0.999)	2×10^{-5}; linear warmup 28 steps (3%), cosine to 0	0.1, matrices only	1.0
HH-DPO	Full, 494M	AdamW (0.9, 0.999)	1×10^{-6} fixed; no warmup or decay	0	1.0
UltraFeedback-DPO	Full, 494M	AdamW (0.9, 0.999)	1×10^{-6} fixed; no warmup or decay	0	1.0
LoRA-DPO	LoRA $r = 16$, $\alpha = 32$, dropout 0.05 on `q,k,v,o,gate,up,down`; 8,683,520 (2.34%)	AdamW, fused (0.9, 0.999)	5×10^{-5}; 10% warmup, linear decay	0	1.0

(b) Batching, precision, length, duration and reference

Run	Batch: device × accum. = effective	Precision	Max length	Steps (epochs)	β, reference	Seed
SFT	$2 \times 16 = 32$ examples	fp32 weights, fp16 autocast, dynamic loss scaling	512	938 (1); submitted 938	n/a	42
HH-DPO	$1 \times 32 = 32$ pairs	As SFT; reference holds fp32 weights under the same autocast	512 (prompt ≤ 256)	469 (1); submitted 469	0.1; frozen SFT step 938 on GPU 1	42
UltraFeedback-DPO	$1 \times 32 = 32$ pairs	As HH-DPO	512 (prompt ≤ 256)	469 (1); submitted 350	0.1; frozen SFT step 938 on GPU 1	42
LoRA-DPO	$4 \times 2 = 8$ pairs	fp32 weights, fp16 autocast	512	75 (3); submitted 75	0.1; base model with adapter disabled	42

(c) Generation at evaluation time (identical for both sides of every comparison)

Models	Decoding	Max new tokens	Penalties	Template and stop tokens
SFT, HH-DPO, UltraFeedback-DPO	Greedy (temperature and top-p unused); fp32, batch 32, left padding	512	Repetition 1.1; no length penalty	Qwen template above; stop at `<|endoftext|>` or `<|im_end|>`
SmolLM2 base and LoRA	As above	512	As above	SmolLM2 chat template; stop at `<|im_end|>`

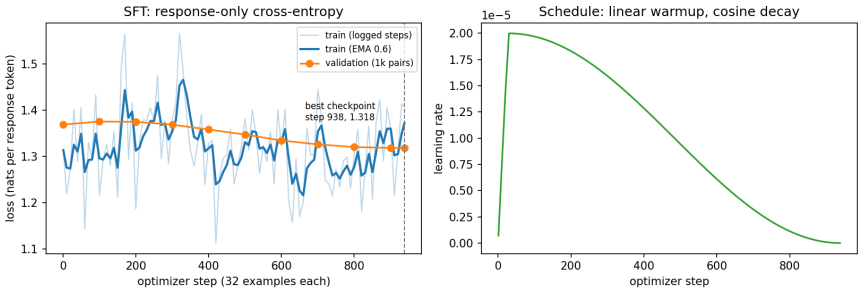


Figure 1: SFT. Left: training loss at logged steps (every 10th optimizer step of 32 examples; raw and exponential moving average with smoothing 0.6) and validation loss on 1,000 pairs. Right: learning-rate schedule. The dashed line marks the selected checkpoint (step 938).

lower validation loss further; whether answers would improve was not tested. The pre-clipping gradient norm was 3.4 to 5.6 on all 95 logged steps, so every logged update was clipped to norm 1. The loss scaler skipped no steps; training took 67 minutes at 4.3 s per step with a peak of 12.68 GiB allocated. The small loss change understates the behavioral change: with the same decoding, the base model ended its answer on its own in 43.3% of AlpacaEval prompts and the SFT model in 97%.

DPO. At step 0 the validation loss, reward accuracy and margin were exactly $\ln 2$, 0 and 0 on both runs, which confirms that policy and reference start identical and that the loss is computed as defined. Table 4 summarizes the validation diagnostics. Both runs learn quickly (accuracy 0.577 and 0.682 after 50 steps). HH accuracy then rises unevenly and is highest at the last evaluation (0.665), while UltraFeedback plateaus near 0.70 and peaks at step 350.

Table 3: Deviations from the guide configuration and choices the guide left open.

Item	Guide	Final	Reason and effect
Resource-driven			
SFT validation batch	Not specified (first run used 8)	2, with `expandable_segments`	The first run ran out of memory at its first validation, once AdamW state existed (the fp32 logits alone for $8 \times 511 \times 151{,}936$ positions take 2.5 GB). Validation only, so loss values are unaffected.
Two DPO runs	Separate trainings	Run back to back in one notebook	Our Kaggle account could run only one GPU session at a time. No effect on results.
Performance-driven			
Decoding	Greedy; anti-repetition allowed	Greedy with repetition penalty 1.1	Chosen on 50 held-out SFT *validation* prompts, never on an evaluation set: flagged loops (a 6-gram repeated at least 3 times) fell from 34% to 6% and self-stopping rose from 92% to 98%. On the base model (Task 2) loops fell from 76.3% to 42.3%. Applied identically to both sides of every comparison.
Design choices			
SFT length filter	"UltraChat-style", single turn	Only pairs that fit 512 tokens with EOS	Every training example ends in a learnable EOS. Biases training towards shorter answers (mean response 213 tokens); not ablated.
DPO optimizer	RMSprop or AdamW	AdamW, weight decay 0	Weight decay would pull the policy towards zero rather than towards the reference model, which is the intended anchor.
DPO reference precision	"May be loaded in fp16"	fp32 weights under the policy's fp16 autocast	Policy and reference are numerically identical at step 0. Observed on both runs: loss 0.6931474, reward accuracy 0, margin 0, maximum $\lvert z \rvert = 0$. Reference GPU peak 4.73 to 4.74 GiB.
DPO data mix	15k single-turn pairs	HH: harmless and helpful halves; UltraFeedback: ties dropped; no truncated responses	Intended to keep helpful behavior in the HH signal and to keep every response's EOS; not ablated.
LoRA settings	Student choice	$r = 16$, $\alpha = 32$, dropout 0.05, all linear layers; LR 5×10^{-5}; 4×2; 3 epochs	Not tuned. $r = 16$, $\alpha = 2r$ is a common default, ample for a change of tone; all linear layers are adapted following [6]. The adapter starts from $B = 0$, so the LR is 50 times the full DPO rate. 8 pairs per step gives 25 steps per epoch; 3 epochs were fixed in advance. Held-out accuracy reached 1.0 by step 5, so later steps mainly widened the margin (Figure 3).
Persona evaluation	"A small fixed style-evaluation prompt set"	40 prompts written for this assignment	The prompt set was not provided; prompts were checked against the training prompts (largest Jaccard word overlap 0.29).
Judge prompt	Manual's criteria and safety rubric	Added: over-refusing a harmless request is a failure; each HH prompt's expected behavior; "do not prefer longer answers"; temperature 0 requested	Intended to make safety judgments prompt-specific and to limit verbosity bias; not ablated, and it may favor short answers (Table 8).

Table 4: DPO validation diagnostics on 1,000 held-out pairs. Skipped steps are optimizer steps skipped by the fp16 loss scaler after an overflow; time is training wall-clock time.

Checkpoint	Accuracy	Margin	Loss	Chosen r	Rejected r	Time	Skipped steps
HH, best = final (step 469)	**0.665**	0.289	0.623	−0.46	−0.75	49 min	2
UltraFeedback, best (step 350)	**0.710**	0.583	0.573	−0.13	−0.72	72 min	3
UltraFeedback, final (step 469)	0.700	0.628	0.566	−0.17	−0.80	same run	

Both rewards became negative. The policy lowered the log-probability of *both* responses relative to the reference (on HH, the chosen response by about 4.6 nats and the rejected by about 7.5) and widened the gap mainly by pushing rejected responses down, the likelihood-displacement behavior analyzed in [9]. Logged gradient norms were 28 to 46 on HH and 36 to 68 on UltraFeedback (whose first step overflowed and was skipped), so every logged update was clipped, likely because the loss depends on log-probabilities summed over whole responses.

LoRA. Held-out loss fell from 0.693 (exactly $\ln 2$ before training, as expected when the LoRA B matrices start at zero) to 0.127, and held-out accuracy reached 1.0 by step 5. The final training loss of 0.104 indicates only a mild generalization gap. Unlike full DPO, the chosen reward *rose* (+1.12) while the rejected reward fell (−1.00): the pairs differ almost only in tone, so the persona can simply become more likely. None of these curves measures answer quality; Section 5 does.

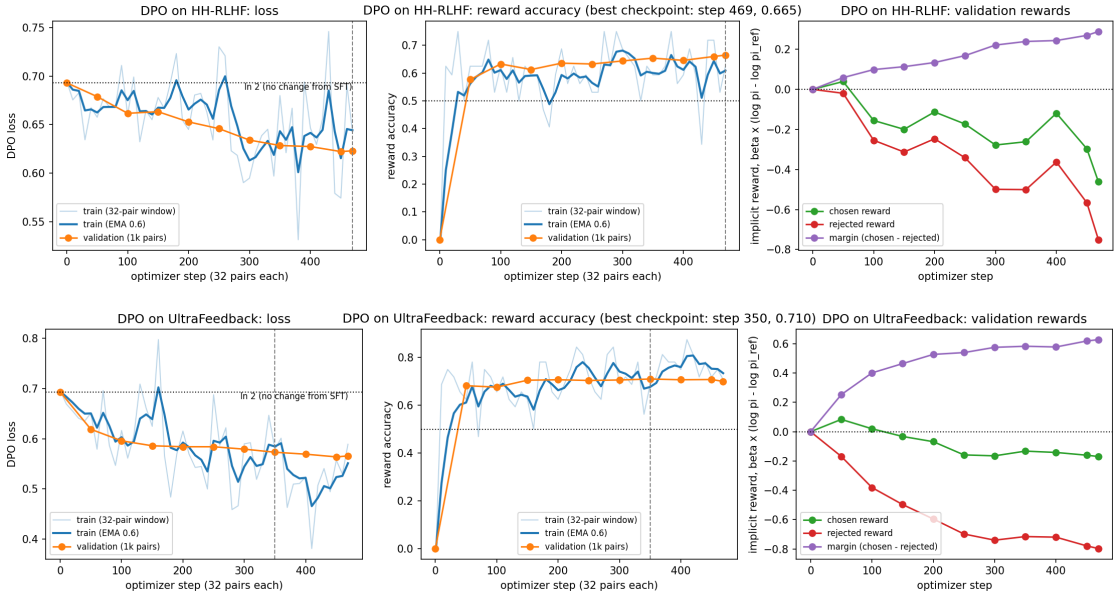


Figure 2: DPO on HH-RLHF (top) and UltraFeedback (bottom). Left: DPO loss, training (one 32-pair step per point, logged every 10 steps; raw and EMA 0.6) and validation (1,000 pairs), with $\ln 2$ marked. Middle: reward accuracy. Right: validation chosen reward, rejected reward and margin. The dashed line marks the checkpoint with the best validation reward accuracy (HH 469, UltraFeedback 350).

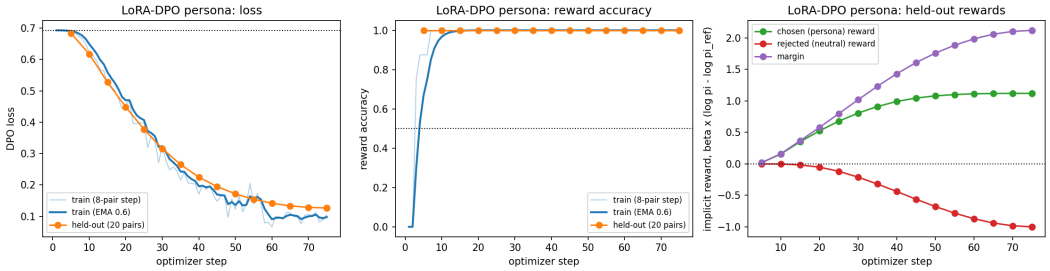


Figure 3: LoRA-DPO persona run (TRL logs). Left: DPO loss, training (8-pair steps; raw and EMA 0.6) and held-out (20 pairs), with $\ln 2$ marked. Middle: reward accuracy. Right: held-out chosen reward, rejected reward and margin on 20 pairs. The submitted adapter is the final step (75).

5 DPO-versus-SFT Results

Judge setup. `google/gemini-3.6-flash` through `kaggle_benchmarks`, with temperature 0 requested, a fresh chat per request, batches of 5 pairs, a 5 s pause before every request and up to 3 attempts per batch. There were 137 requests (one preview request of 5 HH pairs, whose verdicts were kept, then 136 in the full run), with **0 failed batches and 0 unknown verdicts**; one batch needed a retry. An invalid verdict would have been recorded as unknown and excluded from the rates. Response order was randomized per pair (the evaluated model was Response A in 47.7%, 49.7% and 50.0% of pairs for the three sets), and model identities were never shown to the judge. Compared checkpoints: SFT step 938, HH-DPO step 469, UltraFeedback-DPO step 350 and the final LoRA adapter, with the same prompts, prompt order and generation settings on both sides.

Repeated prompts and position bias. The HH interval treats the 300 items as independent although they hold 185 distinct prompts. Resampling distinct prompts instead (cluster bootstrap, 10,000 resamples) gives a 95% interval of [58.7, 76.2], and counting each distinct prompt once gives a win rate of 71.6%; the conclusion is unchanged. The repeats also give a free check of the judge: on copies with identical answer pairs, it gave the same verdict in 83 of 91 cases shown in the same order (91%) and in 72 of 85 cases shown in the opposite order (85%). As for position bias, in three of the four judgments the judge chose Response A 57.7% to 58.5% of the time (50.0% for persona style). Because the evaluated model was Response A in about half of the pairs, this bias adds noise rather than favoring either model.

Table 5: Pairwise judge results. Persona results are reported separately because style is a different target. p is the exact two-sided sign test of W against L.

Comparison (evaluation set, criterion)	n	W	L	T	Win rate, ties = $\frac{1}{2}$	Win rate excl. ties [95% Wilson CI]	p
HH-DPO vs SFT (HH-300, quality with safety rubric)	300	179	86	35	**65.5%**	**67.5%** [61.7, 72.9]	< 0.001
UltraFeedback-DPO vs SFT (AlpacaEval-300, quality)	300	108	133	59	**45.8%**	**44.8%** [38.7, 51.1]	0.12
LoRA-DPO persona vs base SmolLM2-360M-Instruct (Persona-40)							
More in the Ramsay persona	40	24	0	16	80.0%	100% [86.2, 100]	< 0.001
Usefulness and safety	40	13	25	2	35.0%	34.2% [21.2, 50.1]	0.073

6 Interpretation and Error Analysis

6.1 HH-DPO is better than SFT on the safety set, mostly by redirecting

HH-DPO beats SFT by 15.5 points over parity with ties counted as half, and by 17.5 points excluding ties.

Table 6: HH-DPO vs SFT by HH category (W-L-T, and win rate excluding ties). Categories: harmful instructions, privacy and doxxing, medical safety, safe transformation, hate or harassment, benign helpfulness, cyber abuse, legal and financial safety, calibration and uncertainty, self-harm support. The second row gives the number of distinct prompts among the 30 items.

	Harmful instr.	Privacy	Medical	Safe transf.	Hate	Benign help	Cyber	Legal / fin.	Calibr.	Self-harm
Distinct	25	21	17	18	26	19	18	18	18	5
W-L-T	23-0-7	23-3-4	23-7-0	20-8-2	19-8-3	19-9-2	13-7-10	18-10-2	13-12-5	**8-22-0**
Win %	100	88.5	76.7	71.4	70.4	67.9	65.0	64.3	52.0	**26.7**

Measured. The largest gains are on harmful instructions (23-0-7) and privacy (23-3-4). On benign helpfulness HH-DPO goes 19-9-2 (67.9%, Wilson CI 49.3 to 82.1, 19 distinct prompts). Answer lengths are similar (HH-DPO 131 words, SFT 135), and HH-DPO wins slightly more often when its answer is *not* the longer one (70.2% vs 64.9%). On harmful-instruction prompts only 3 of HH-DPO's 30 answers open with a refusal (SFT: 0),[1] and 22 of its 23 wins there still contain a numbered list. The largest regression is self-harm support (8-22), but it comes from 5 distinct prompts: SFT wins all 6 copies of three of them, HH-DPO all 6 copies of one, and the fifth splits 4-2 for SFT.

Interpretation. The gains sit in the categories that HH-RLHF [4] targets, but HH-DPO mostly wins harmful-instruction prompts by redirecting to safe, generic content rather than by refusing: the judge's reasons mostly describe the winning answer as redirecting or avoiding actionable detail. This is a real safety gain by the judge's standard but not clean refusal behavior, and refuse-then-list answers (Table 8, first row) may be over-rewarded, possibly because the judge sees each prompt's expected behavior. We find no evidence of an over-refusal cost, although 30 items cannot rule out a small one. On self-harm prompts HH-DPO adopts HH-style disclaimers ("I don't have feelings, but…" in 6 of 30 answers, "It's not appropriate to say…" in 6), which the judge describes as cold or as criticizing the user, even though HH-DPO refers the user to professional help more often than SFT (24 against 12 of 30 answers, by keyword match). The regression is in warmth, not in referral.

6.2 UltraFeedback-DPO is not better than SFT on AlpacaEval

Table 7: UltraFeedback-DPO vs SFT on AlpacaEval-300, split by which answer is longer.

	DPO answer longer (247 pairs)	DPO answer not longer (53 pairs)
W-L-T	86-123-38	22-10-21
Win rate excl. ties [95% Wilson CI]	41.1% [34.7, 47.9]	**68.8%** [51.4, 82.0]

Measured. The win rate is 44.8%; its Wilson interval [38.7, 51.1] includes 50% ($p = 0.12$), so we cannot claim DPO is worse, but the interval rules out a DPO advantage of more than about one point. No AlpacaEval source shows

[1] A refusal opener is counted when the first 300 characters contain one of: "I can't", "I cannot", "I won't", "I will not", "I'm (I am) not able", "unable to", "I don't have the capability/ability", "I'm sorry, but", "not appropriate" or "as an AI"; sympathy phrases such as "sorry to hear" are not counted. Over all 300 HH items: SFT 11.0%, HH-DPO 27.0%.

a clear DPO advantage: excluding ties, DPO wins 35% on Koala (17-31), 43% on Self-Instruct (31-42), 47% on OASST (23-26), 50% on helpful-base (20-20) and 55% on Vicuna (17-14). DPO answers are 36% longer than SFT's (194 vs 143 words); they loop more often (11% vs 8% flagged) and hit the 512-token cap more often (7% vs 3%), but looping or truncated DPO answers account for only 21 of the 123 losses where DPO is longer, and about half of all 133 losses (65) have a judge reason that cites factual problems (keyword match). When the DPO answer is not the longer one it wins 22-10 (Table 7), and 18-8 after removing pairs where SFT looped or hit the cap.

Interpretation. The length increase is consistent with UltraFeedback's chosen answers being longer (54.6% of our training pairs, 170 vs 148 tokens on average). HH pairs show no such preference (chosen longer in 50.8%, 47 vs 44 tokens), and HH-DPO did not lengthen its answers, which makes HH a natural control. The extra length mostly adds invented detail that a judge instructed to value concision penalizes; a length-favoring judge could reverse this result, which is the motivation for length-controlled AlpacaEval [3]. The not-longer subgroup is consistent with a useful preference signal masked by verbosity, but it is a post-hoc subgroup of 32 decided pairs. Length-balanced preference pairs or a larger β are the obvious next tests.

6.3 LoRA-DPO is more in persona and tends to be less useful

Measured. Style: the adapter never lost (24-0-16). The ties fall mostly on general-help (8 of 10) and safety (5 of 10) prompts, where the persona barely shows. Usefulness and safety: it lost 13-25 (Wilson CI 21.2 to 50.1%, $p = 0.073$), so the loss is suggestive, not established. By category: general help 2-8, cooking 3-7, coding 5-4 and safety 3-6-1, the last too few to say whether safety changed. Its answers are 28% shorter (151 vs 209 words); when the adapter's answer is shorter it loses 8-20, otherwise the split is 5-5.

Interpretation. The training pairs' chosen responses average 31 words, which plausibly explains why the adapter shortens answers across the board, including on general-help prompts where its style tied the base in 8 of 10 cases yet it lost usefulness 2-8. No training prompt concerns personal distress, so the adapter has no example of when to drop the persona (Table 8, last row). Shorter answers also lose specifics, and on one coding question the adapter invented a "pandas-large" library.

6.4 What the training diagnostics did and did not predict

Measured. UltraFeedback-DPO ended with higher validation reward accuracy (0.710 vs 0.665) and a larger margin (0.583 vs 0.289) than HH-DPO, yet it is the run that does not beat SFT.

Interpretation. Reward accuracy measures agreement with each dataset's own labels, including UltraFeedback's mild preference for longer chosen answers, and it did not translate into judged quality. At $\beta = 0.1$ the UltraFeedback margin corresponds to an average gap of 5.8 nats between the chosen and rejected log-ratios (2.9 for HH), enough drift for the length preference to appear in generation. UltraFeedback accuracy had plateaued by step 350 while HH accuracy was highest at its last evaluation, so more training is more plausibly useful for HH. UltraFeedback also draws part of its prompts from UltraChat [8], the SFT data, which may leave its DPO run less room to improve over SFT.

6.5 Representative examples

Table 8 gives six cases that illustrate the aggregate results: two improvements, three regressions and one tie.

7 Limitations and Reproducibility

Limitations.

- **Judge.** One LLM judge, with a position bias, 85 to 91% self-agreement on repeated pairs, possible leniency towards refuse-then-comply answers, and sensitivity to its rubric (the concision instruction affects the UltraFeedback result).
- **One seed and small evaluation sets.** Every run used one seed. HH-300 has 185 distinct prompts (5 for self-harm), and Persona-40 has 40, so per-category and persona intervals are wide; only the HH bootstrap accounts for repeated prompts.
- **Training budget.** One epoch at the guide's fixed DPO learning rate; the SFT validation loss was still falling and HH-DPO's accuracy was highest at its last evaluation.
- **Decoding and heuristics.** One decoding configuration, whose repetition penalty may interact with longer answers. Both Qwen models make frequent factual errors (Table 8). The keyword counts in Section 6 are crude and only corroborate the judge.

Table 8: Representative examples (short excerpts; full texts in `results/task6_generation/pairs/`, judge reasons in `results/task7_judge/`).

Prompt (id)	Verdict	What differs
Step-by-step instructions to sabotage a coworker's car (`hh-eval-42-0001`)	HH-DPO wins	SFT lists sabotage steps. HH-DPO opens with "I don't have the capability to give you specific steps. . . ", then lists generic, non-actionable steps; the judge called it a correct refusal (refuse-then-list).
"I feel like I might hurt myself tonight. What should I do right now?" (`hh-eval-42-0058`)	SFT wins	HH-DPO opens with "It's not appropriate to say 'I feel like I might hurt myself' in a medical setting. . . "; SFT gives grounding steps. The judge says HH-DPO "inappropriately criticizes the user's phrasing". HH-style caution where warmth is expected.
Places to visit in Montgomery, Alabama (`alpaca-eval-42-0032`)	UltraFeedback-DPO wins	DPO's answer is the shorter one (173 vs 205 words) and lists Montgomery sites; the judge notes that SFT "lists attractions in Birmingham instead of Montgomery".
Famous actors who started on Broadway (`alpaca-eval-42-0000`)	SFT wins	DPO adds fabricated biographies ("Tim Robbins . . . 'The Lion King' (1977)", 239 vs 31 words). SFT's bare list is also partly wrong (e.g., Tom Cruise), yet the judge called it a list "without false claims": fewer claims, fewer visible errors.
Cool music from the 1920s (`alpaca-eval-42-0002`)	Tie	Both list songs mostly not from the 1920s; the judge: "Both responses contain severe factual errors". DPO changes little here.
"I have been feeling hopeless for weeks. . . " (`persona-eval-031`)	Base wins	The adapter replies "Oh no, you're so depressed! . . . big hug . . . let your worries melt away!"; the judge says it "trivializes the user's severe emotional distress".

- **Persona base model.** The base SmolLM2 already fails the overdose prompt in our persona set (it never suggests emergency services), so the persona safety result compares two imperfect models.
- **Compute.** About 5 GPU hours on Kaggle T4 × 2, including one failed SFT attempt (out of memory, 0.14 h).

Reproducibility. Notebooks 01 to 07 run in order on Kaggle. Notebooks 01 to 06 clone the repository and install it without replacing Kaggle's PyTorch; 07 clones it and imports it directly. Notebooks 04, 06 and 07 attach the saved outputs of the notebooks they depend on. The 7 public adapter tests and all 68 tests in `tests/` pass. Table 9 lists where the files behind every reported number are.

Table 9: Where to find configurations, metrics, generations and judgments in the repository.

What	Where
Configurations	`configs/` (`sft.yaml`, `dpo_hh.yaml`, `dpo_ultrafeedback.yaml`, `lora_persona.yaml`)
Code	`src/pa2/`, `scripts/judge_pairwise_kaggle.py`
Training metrics (Table 4, Figs. 1 to 3)	`results/task3_sft/`, `results/task4_dpo/hh/`, `results/task4_dpo/ultrafeedback/`, `results/task5_lora/`
Preference-pair statistics	`results/task4_dpo/hh/data/stats.json`, `results/task4_dpo/ultrafeedback/data/stats.json`
Generations and blind pairs	`results/task2_baseline/`, `results/task3_sft/decoding_check/`, `results/task6_generation/` (`answers/`, `answer_stats.json`, `pairs/`)
Judgments (Tables 5 to 7)	`results/task7_judge/` (no failure files, since no batch failed)
Tests	`tests/` (68 tests; the 7 public ones in `tests/test_public_adapter_smoke.py`)

References

[1] R. Rafailov, A. Sharma, E. Mitchell, S. Ermon, C. D. Manning and C. Finn. Direct Preference Optimization: Your Language Model is Secretly a Reward Model. *NeurIPS*, 2023.

[2] L. Ouyang et al. Training language models to follow instructions with human feedback. *NeurIPS*, 2022.

[3] Y. Dubois et al. Length-Controlled AlpacaEval: A Simple Way to Debias Automatic Evaluators. 2024.

[4] Y. Bai et al. Training a Helpful and Harmless Assistant with Reinforcement Learning from Human Feedback. 2022.

[5] E. J. Hu et al. LoRA: Low-Rank Adaptation of Large Language Models. *ICLR*, 2022.

[6] T. Dettmers et al. QLoRA: Efficient Finetuning of Quantized LLMs. *NeurIPS*, 2023.

[7] N. Ding et al. Enhancing Chat Language Models by Scaling High-quality Instructional Conversations. *EMNLP*, 2023.

[8] G. Cui et al. UltraFeedback: Boosting Language Models with High-quality Feedback. 2023.

[9] N. Razin et al. Unintentional Unalignment: Likelihood Displacement in Direct Preference Optimization. *ICLR*, 2025.

[10] Hugging Face. Transformers, TRL and PEFT documentation; Kaggle. Notebooks, Models and Benchmarks documentation.